ASEM Talint

Dashboard-first semiconductor talent decision intelligence

Deterministic readiness scoring plus Z.AI GLM structured reasoning

Built for training-path fit, evidence review, and market-context visibility

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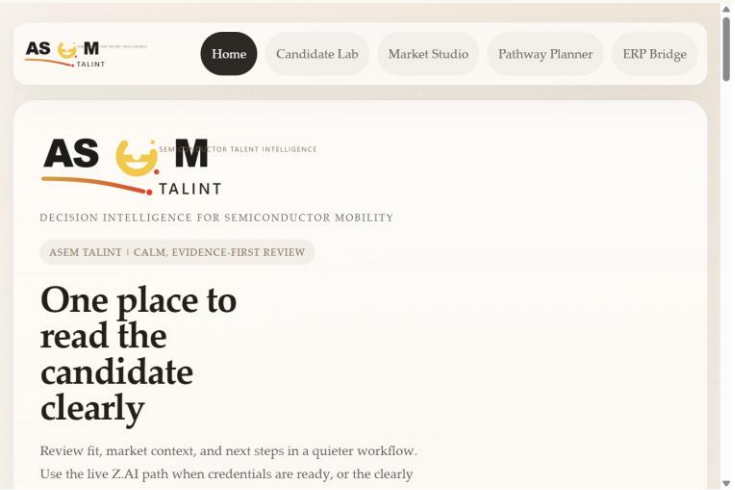
One place to read the candidate clearly, with live market context and auditable AI explanation.



Problem: placement decisions are still fragmented

What breaks today

Candidate-to-track decisions are still reviewed manually and inconsistently.

Missing inputs, access constraints, and labor-market context are often separated.

Pathway decisions should connect training to real economic outcomes, not only admissions workflow.

Why it matters

Semiconductor pathways require scarce seats, scarce OJT capacity, and better matching discipline.

Reviewers need auditable evidence, not a generic chat response.

Malaysia's industrial push makes local-value pathway allocation more strategic.

Our response

Standardize candidate evidence.

Combine deterministic scoring with official data slices.

Use Z.AI GLM for structured explanation on the judged path.

Citations: UMHackathon Domain 2 brief; UMHackathon 2026 Judging Criteria

Economic theory: what friction we reduce

Human capital allocation

Training seats and OJT slots are scarce investments.

Signaling and information asymmetry

Resumes and credentials are noisy until structured.

Search and matching frictions

Candidate, track, and employer fit must be aligned.

Spatial and wage frictions

Access and economic payoff change pathway quality.

Feature mapping: Candidate Lab -> signals | Market Studio/OJT -> matching | GIS accessibility -> spatial friction | Wage mobility -> economic payoff | ERP Bridge -> coordination cost

Economic framing source: submission/preliminary/economic-theory-anchor.md

Why this matters in Malaysia now

"Made by Malaysia" raises the bar for local value creation.

Talent pathways need stronger links to vendors, SMEs, and industry demand.

Training decisions should be explainable, evidence-backed, and fast.

A pathway is stronger when access, wage direction, and employer demand all support it.

Malaysia lens

Local capability, not just placement

The national push is toward stronger local vendors, deeper SME participation, and tighter links between investment, training, and research. ASEM Talint keeps that lens visible while pathway decisions are being made.

NST source, 5 Feb 2026

Citation: New Straits Times, 5 Feb 2026, 'Shift to Made by Malaysia strategy'

Users and decision workflow

concrete.

Primary user

Training reviewer or program lead

Needs fit, gaps, readiness, and next action in one place

Supporting user

Employer-partnership or OJT reviewer

Needs match quality, access, and wage context

1. Ingest candidate and track evidence

2. Parse resume and standardize signals

3. Compute deterministic fit and market context

4. Use Z.AI GLM to generate structured explanation

Workflow source: PRD, pitch deck outline, and current live prototype flow

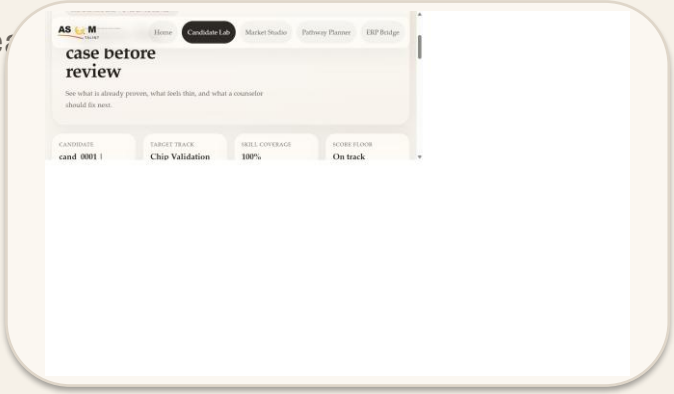
MVP scope and product surfaces

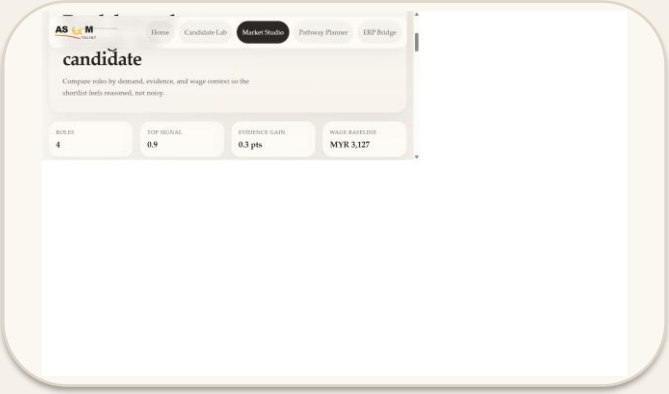
In scope now

One candidate-to-track fit workflow

Resume parsing for PDF and DOCX with OCR fallback

Wages, employer demand, accessibility, OJT, and wage-mobility panels



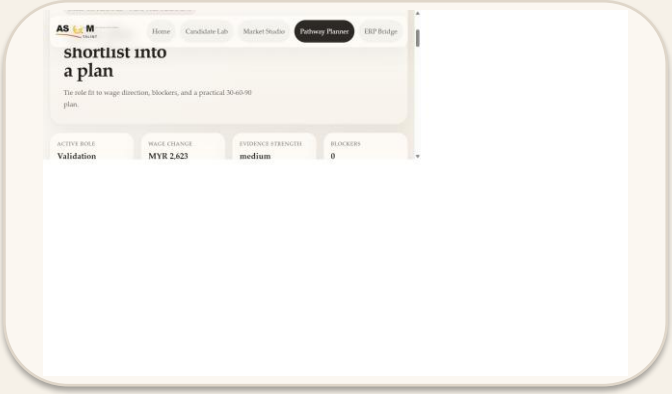


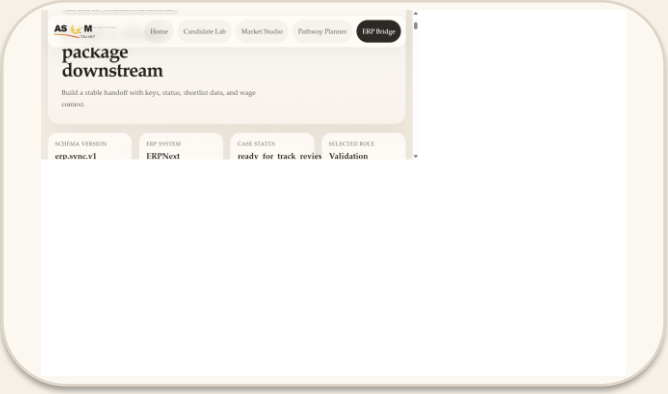
Still out of scope

Production authentication and reviewer case management

Full persistence and operational analytics

Public deployment claims beyond local validation





Snapshots: Candidate Lab, Market Studio, Pathway Planner, ERP Bridge from current local app

Why Z.AI GLM is central

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Deterministic context
Scoring, thresholds, missing-input handling, and wage/access logic remain in code.

Prompt builder
Bounded JSON evidence is passed forward so the model explains a structured decision problem.

Z.AI GLM provider
This is the judged reasoning path in the shipped product flow.

Schema validation
Model output must parse into structured response JSON before the app uses it.

Dashboard/API consumer
Reviewers see recommendation, tradeoffs, missing inputs, and pathway steps.

No silent model replacement: the optional ILMU route is compatibility infrastructure, not the judged runtime path.

Architecture rule: keep Z.AI GLM central in implementation, narrative, and demo

System architecture and feasibility

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p

FastAPI interface	Dashboard, decision API, resume API, market routes
Decision engine	Deterministic context + provider boundary
Data and services	Resume parsing, wages, hotspots, OJT, wage mobility
Output contracts	Structured JSON, ERP package, observable errors

Typed request and response models keep the interfaces explicit.

Prompt budgets, token budgets, and upload limits are enforced in code.

Official-data slices are normalized locally for preliminary feasibility.

ERP Bridge packages the same decision output into operational sync-ready JSON.

The architecture reduces screening, matching, spatial, and coordination frictions in one workflow.

Feasibility source: SAD, README, and live prototype implementation

Slide 9

Live demo walkthrough

Show the runtime-ready status and the sample candidate payload.

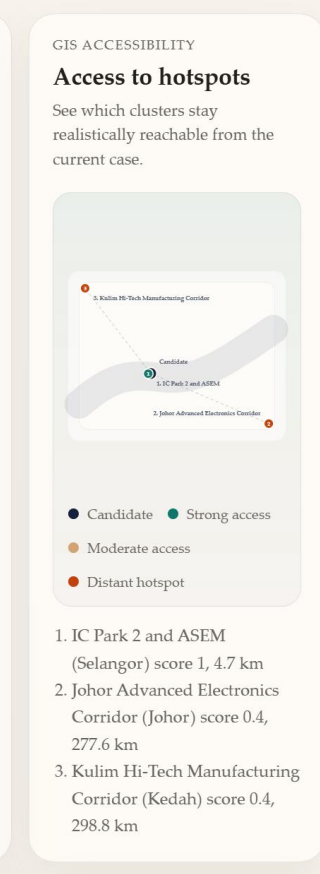
Parse and apply a resume so judges see structured evidence ingestion.

Refresh market panels for wages, employer demand, accessibility, and OJT.

Pause on the cleaned GIS accessibility figure with numbered hotspots and ranked list.

Run the live Z.AI route and inspect recommendation, tradeoffs, and missing inputs.

Open the raw structured response briefly to prove contract discipline.



Fallback still image in deck: cleaned GIS accessibility panel from current dashboard build

Engineering quality and risk control

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Quality controls

Typed models and explicit provider boundaries

Provider output validated before use

Malformed or empty model output fails visibly

Test coverage

Happy path and invalid payload routes

OCR fallback and parser warnings

Health/readiness behavior and judged-path guardrails

Observability

Provider usage metrics surfaced

Warnings kept explicit

No silent degradation of judged model path

Risk excerpt

Live provider readiness	**Medium**	Validate `/health`, fail visibly, no silent substitution
Data refresh discipline	**Medium**	Use bounded official-data slices and document refresh path
Submission packaging quality	**Low-Med**	Keep PRD, SAD, QATD, repo, deck, and video aligned

Validation evidence: pytest route tests, provider parsing, OCR fallback, and documented QATD risk controls

Closing

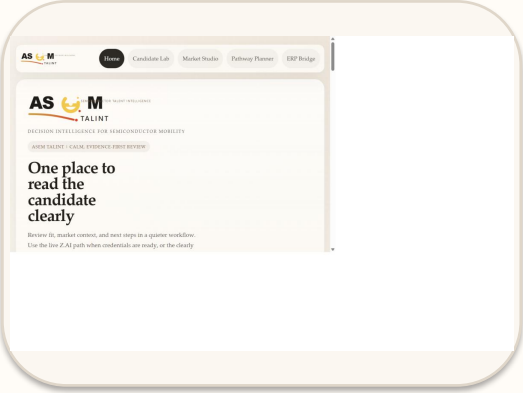
Feasible prototype, honest scope, submission-ready structure

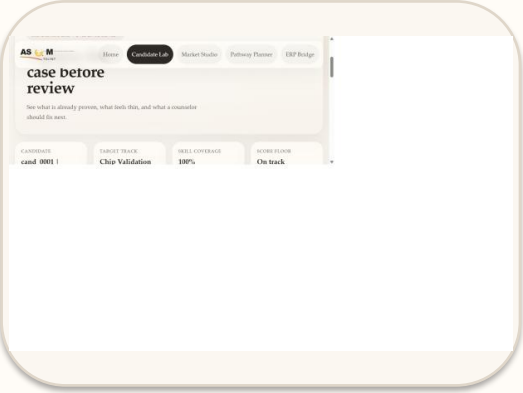
ASEM Talint fits Domain 2 because it supports economically meaningful pathway decisions.

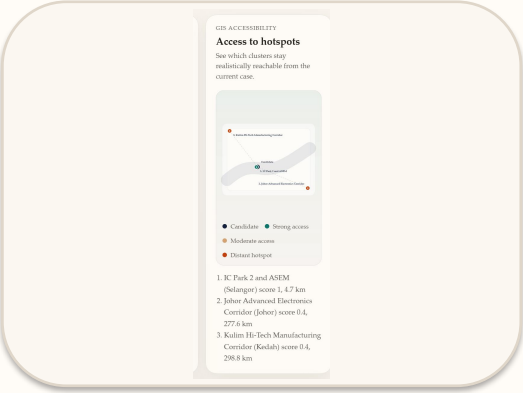
Z.AI GLM remains central in the product, architecture, and demo narrative.

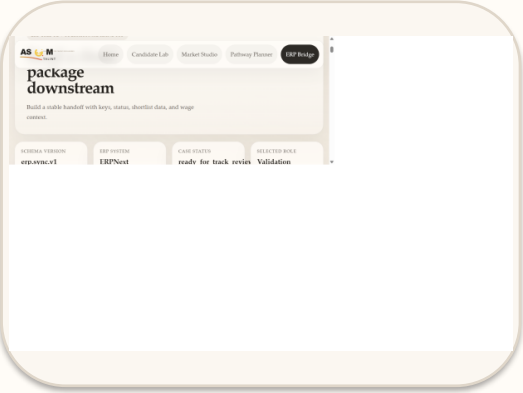
PRD, SAD, QATD, repository, deck, and video runbook are structured for preliminary submission.

Repository: github.com/khaisernong/ASEM-Talint









ASEM Talint turns candidate placement from a manual review task into an auditable, market-aware, Z.AI-centered decision workflow.